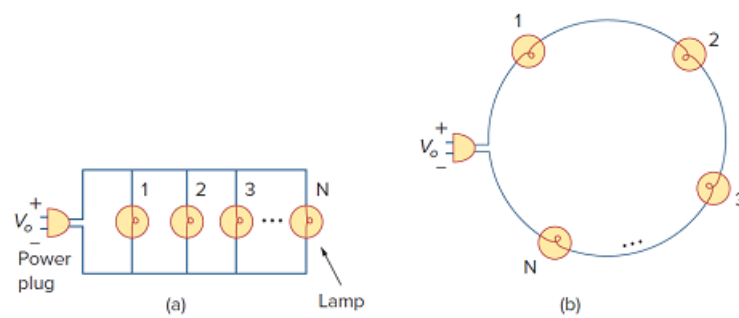


Problem

Refer to Fig. 2.55 and assume there are six light bulbs that can be connected in parallel and six different light bulbs that can be connected in series. In either case, each light bulb is to operate at 40 W. If the voltage at the plug is 115 V for the parallel and series connections, calculate the current through and the voltage across each bulb for both cases.

Figure 2.55: (a) Parallel connection of light bulbs, (b) series connection of light bulbs.



Step-by-step solution

Step 1 of 4

Refer to Figure 2.55 in the textbook.

Case (i):

When 6 light bulbs rated at 40 W are connected in parallel across a 115 V source:

In parallel connection the voltage across each element is same as the source voltage. So, when 6 light bulbs are connected parallel to the source, then all 6 bulbs have the same voltage as source voltage that is 115 V.

Therefore, the voltage across each bulb in parallel connection is **115 V**.

Step 2 of 4

Write the expression to find the power a light bulb.

$$p = VI$$

Use this expression to find the current through the bulb.

$$\begin{aligned} I &= \frac{p}{V} \\ &= \frac{40}{115} \\ &= 0.3478 \text{ A} \\ &= 347.8 \text{ mA} \end{aligned}$$

Therefore, the current through each bulb in parallel connection is **347.8 mA**.

Step 3 of 4

Case (ii):

When 6 light bulbs rated at 40 W are connected in series to a 115 V source:

When light bulbs are connected in series, the total power supplied by the source is the sum of the total power absorbed by the bulbs. Therefore, the total power is,

$$\begin{aligned} p &= (40)(6) \\ &= 240 \text{ W} \end{aligned}$$

Calculate the total current in the circuit.

$$\begin{aligned} I &= \frac{p}{V} \\ &= \frac{240}{115} \\ &= 2.087 \text{ A} \end{aligned}$$

Since, all the 6 light bulbs are connected in series to the source; same source current is flowing through each of the bulbs.

Therefore, the current through each bulb in series connection is 2.087 A.

Step 4 of 4

Calculate the voltage rating of individual bulbs using the power and current rating of the bulbs.

$$\begin{aligned} V &= \frac{p}{I} \\ &= \frac{40}{2.087} \\ &= 19.17 \text{ V} \end{aligned}$$

Therefore, the voltage across each bulb in parallel connection is 19.17 V.